

Dove prism

Y14 (2014) — English translation

In an isosceles rectangular glass prism, the base AB and side face BC are smooth, and the face AC is matte. The base AB of the prism stands on the newspaper. The refractive index of the prism is $n = 1.6$.

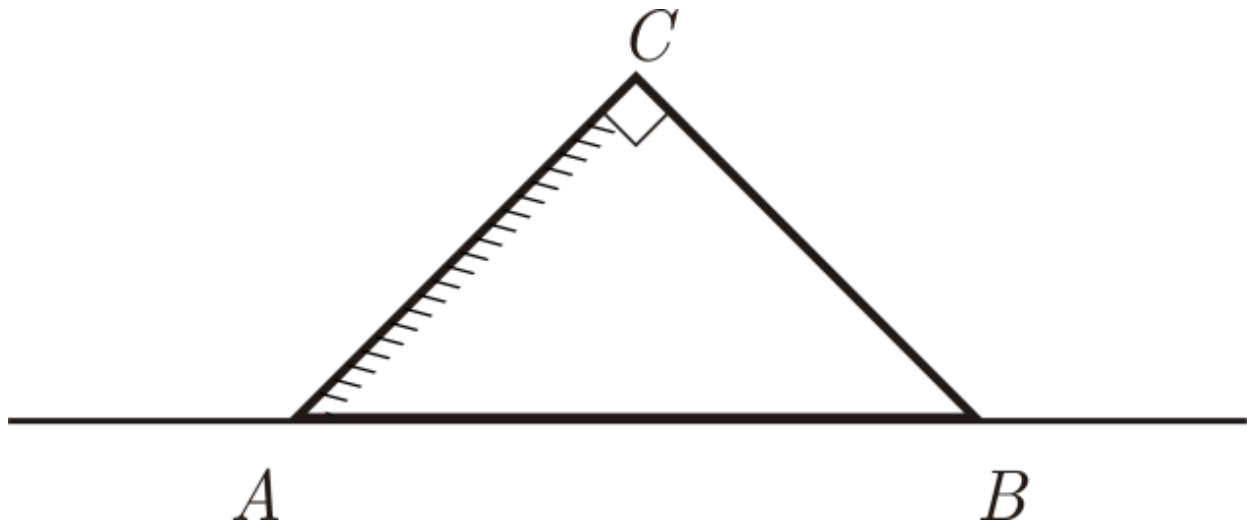


Fig. 1.

A1	What part of the α text is seen by an observer looking at the text through the edge BC ?	0
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The following questions consider a beam of rays incident on a prism parallel to the face AB . To reverse (invert) an image, a Dove prism is often used, which is a truncated rectangular isosceles prism.

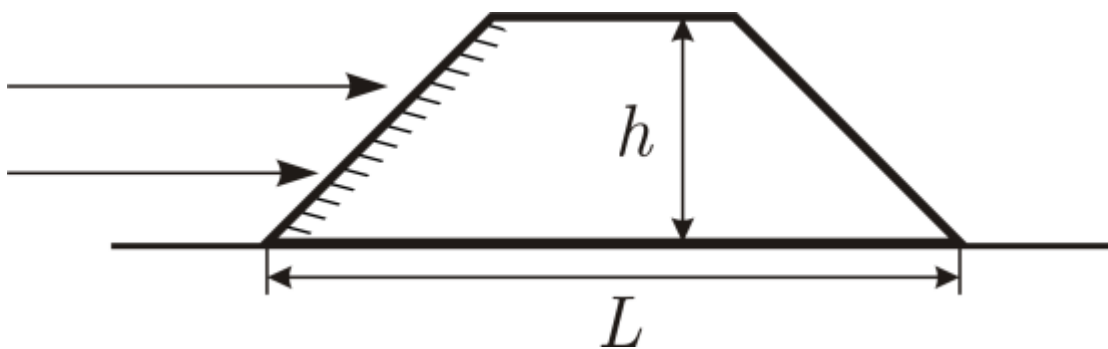


Fig. 2.

A2	Determine the height h of the prism if the length of its base is $L = 10$ cm. The prism should wrap around a beam of maximum cross-section and not contain “non-functioning” glass.	0
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A3	At what values of the refractive index n will no part of the light escape from the face AB ?	0
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Source: <https://pho.rs/p/3966>